

求级数

$$\sum_{n=0}^{\infty} \frac{1}{(4n+1)(4n+2)(4n+3)(4n+4)}$$

注意到

$$\begin{aligned} B(4n+1, 4) &= \int_0^1 x^{4n} (1-x)^3 dx \\ &= \frac{3!}{(4n+1)(4n+2)(4n+3)(4n+4)} \end{aligned}$$

因此

$$\begin{aligned} \sum_{n=0}^{\infty} \frac{1}{(4n+1)(4n+2)(4n+3)(4n+4)} &= \sum_{n=0}^{\infty} \frac{1}{3!} \int_0^1 x^{4n} (1-x)^3 dx \\ &= \frac{1}{6} \int_0^1 \left(\sum_{n=0}^{\infty} x^{4n} \right) (1-x)^3 dx \\ &= \frac{1}{6} \int_0^1 \frac{(1-x)^3}{1-x^4} dx \\ &= \frac{1}{6} \int_0^1 \frac{2}{1+x} - \frac{1+x}{1+x^2} dx \\ &= \frac{1}{4} \ln 2 - \frac{\pi}{24} \end{aligned}$$